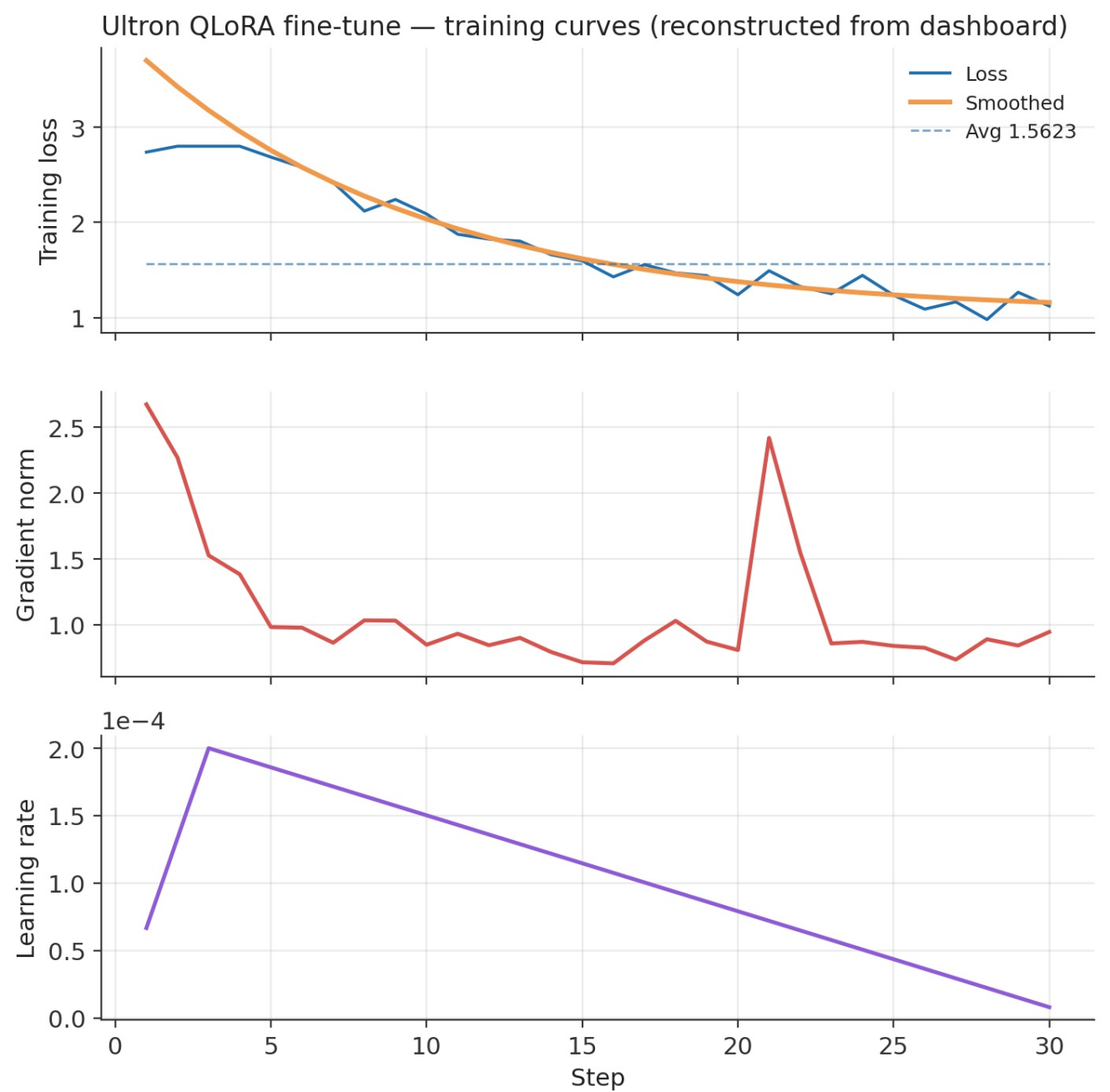


Evaluation Report — Ultron-Qwen3.5-2B-ToolCall

1. What Has Been Measured So Far



Metric	Value
Final training loss	1.2061
Smoothed average loss	1.5623
Final gradient norm	0.993
Steps completed	30 / 30 (100%)
Tokens processed	40,580
Training time	2m 14s (0.22 steps/s)

Full curves (loss, gradient norm, learning-rate schedule) are in the companion fine-tuning report, [Ultron_Qwen3.5-2B_Finetuning_Report.docx](#).

This is training-set fit only. No validation or held-out test data was scored during this run.

2. Gap: No Held-Out Evaluation

The training dashboard explicitly reports "Evaluation not configured." This means:

- We cannot currently distinguish genuine learning from memorization of the training examples.
- We have no measured accuracy figure for correct tool selection, correct argument formatting, or correct refusal behavior on unseen prompts.
- Any claim of "the model got better at tool calling" is currently supported only by the loss curve shape, not by task-level accuracy.

3. Evaluation Plan (to run before/at demo)

3.1 Held-out split

Carve out 10-15% of the custom dataset (not seen during training) and score:

- **Tool-selection accuracy:** did the model call the correct tool for the instruction?
- **Argument accuracy:** exact/partial match of required arguments (engagement_id, target, scan_type, etc.)
- **Format validity:** does the output parse as a valid function call every time?

3.2 Out-of-distribution probe set

A small hand-written set of prompts not resembling training examples, to sanity-check generalization rather than memorization — including:

- Ambiguous instructions that should trigger `request_human_confirmation` rather than a direct action
- Out-of-scope target requests that should be refused/rejected
- Multi-step instructions requiring more than one tool call in sequence

3.3 Comparison to base model

Run the same held-out and probe sets against the un-fine-tuned Qwen3.5-2B base model to quantify the actual lift from fine-tuning, not just the absolute score.

3.4 Standard benchmark (optional, for external comparability)

If time allows, run a BFCL-style function-calling benchmark subset against both the base and fine-tuned model, so results are comparable to the public numbers cited in the model-selection report.

4. Results Table (fill in once run)

Metric	Base Qwen3.5-2B	Fine-tuned (this checkpoint)
Tool-selection accuracy	[TBD]	[TBD]
Argument accuracy	[TBD]	[TBD]
Format validity rate	[TBD]	[TBD]
Correct refusal on out-of-scope prompts	[TBD]	[TBD]
Correct escalation to human confirmation	[TBD]	[TBD]

5. Known Limitations of This Evaluation Plan

- Small dataset size (single 30-step / ~40K token run) limits how large a held-out set can be without starving training further.
- No adversarial/prompt-injection evaluation is included yet (tracked separately in `RESPONSIBLE_USE_POLICY.md`).
- Results from a hand-written probe set are illustrative, not statistically rigorous — treat as a sanity check, not a certification.